Exercise 20.1

See if you can prove that K-means always converges.

Solution:

We show that $\sum_k \sum_{n|r_k^(n)=1} d(m^{(k)}, x^{(n)})$ is a Lyapunov function.
When assigning, the function always decrease since $d(m^{(k)}, x^{(n)}) > d(m'^{(k)}, x^{(n)})$
We need to show the function decreases when we update the mean.
Let $L_k = \sum_{n|r_k^{(n)}=1} d(m^{(k)}, x^{(n)}) = \sum_{n|r_k^(n)=1} \frac{1}{2} \sum_i (m_i^{(k)} - x_i^{(n)})^2$

$$\begin{aligned}\frac{\delta L_k}{\delta m_i} &= \sum_{n|r_k^{(n)}=1} (m_i^{(k)} - x_i^{(n)}) \\ &= N m_i - \sum_{n|r_k^{(n)}=1} x_i^{(n)}\end{aligned}$$

$$\begin{aligned}&\frac{\delta L_k}{\delta m_i} = 0 \\ \iff\ & m_i = \frac{\sum_{n|r_k^{(n)}=1} x_i^{(n)}}{N}\end{aligned}$$

This means L_k is minimised when m_i is the average of its members' x_i. Hence, when we update m to the average of its members', it must decrease from the previous L_k (equal case is eliminated since that means the means converged and the algorithm terminates).
Also, the original function must be greater than or equal to 0 (equality when $\leq k$ unique points), so from those, the original function decreases. This proves that the K-means always converges.

Exercise 20.2

Show that as the stiffness β goes to ∞, the soft K-means algorithm becomes identical to the original hard K-means algorithm, except for the way in which means with no assigned points behave. Describe what those means do instead of sitting still.

Solution:

For $r_k^{(n)} = 1$,

$$\begin{aligned}
&\frac{\exp(-\beta d(m^{(k)}, x^{(n)}))}{\sum_{k'} \exp(-\beta d(m^{(k')}, x^{(n)}))} \\
&= \frac{1}{1 + \sum_{k' \neq k} \exp(\beta(d(m^{(k)}, x^{(n)}) - d(m^{(k')}, x^{(n)})))} \\
&\to 1
\end{aligned}$$

This is because as $\beta \to \infty$, $\sum_{k' \neq k} \exp(\beta(d(m^{(k)}, x^{(n)}) - d(m^{(k')}, x^{(n)}))) \to 0$, since $d(m^{(k)}, x^{(n)}) < d(m^{(k')}, x^{(n)})$ for all $k' \neq k$.

However, for $r_k^{(n)} = 0$, let $k^* = k^*(n)$ be the mean closest to $x^{(n)}$, and write $d_j = d(m^{(j)}, x^{(n)})$. Then

$$\begin{aligned}
&\frac{\exp(-\beta d(m^{(k)}, x^{(n)}))}{\sum_{k'} \exp(-\beta d(m^{(k')}, x^{(n)}))} \\
&= \frac{1}{1 + \exp(\beta(d_k - d_{k^*})) + \sum_{k' \neq k, k^*} \exp(\beta(d_k - d_{k'}))} \\
&\to 0
\end{aligned}$$

This is because as $\beta \to \infty$,

$$\exp(\beta(d(m^{(k)}, x^{(n)}) - d(m^{(k^*)}, x^{(n)}))) \to \infty$$

and

$$\sum_{k' \neq k, k^*} \exp(\beta(d(m^{(k)}, x^{(n)}) - d(m^{(k')}, x^{(n)}))) \geq 0.$$

Above shows that soft K-means is identical to hard K-means when $\beta \to \infty$.

Now take a mean k with no assigned points, so $r_k^{(n)} \to 0$ for every n and its update is of the form $0/0$. Let $Z_n = \sum_{k'} \exp(-\beta d(m^{(k')}, x^{(n)})) \approx \exp(-\beta d(m^{(k^*)}, x^{(n)}))$, so $r_k^{(n)} = \exp(-\beta d(m^{(k)}, x^{(n)}))/Z_n \approx \exp(-\beta \Delta^{(n)})$, where $\Delta^{(n)} = d(m^{(k)}, x^{(n)}) - d(m^{(k^*)}, x^{(n)}) > 0$.
Let $n^* = \arg\min_n \Delta^{(n)}$. Dividing the top and bottom by $\exp(-\beta \Delta^{(n^*)})$,

$$\frac{\sum_n r_k^{(n)} x^{(n)}}{\sum_n r_k^{(n)}} \approx \frac{x^{(n^*)} + \sum_{n \neq n^*} \exp(-\beta(\Delta^{(n)} - \Delta^{(n^*)})) x^{(n)}}{1 + \sum_{n \neq n^*} \exp(-\beta(\Delta^{(n)} - \Delta^{(n^*)}))} \to x^{(n^*)}$$

since $\Delta^{(n)} - \Delta^{(n^*)} > 0$ for $n \neq n^*$.
This means a mean with no assigned points moves onto the point it lost by the smallest margin ($\Delta^{(n)}$ minimum). If several points tie, it moves to their average.

Exercise 20.3

Explore the properties of the soft K-means algorithm, version 1, assuming that the data points $\{x\}$ come from a single separable two-dimensional Gaussian distribution with mean zero and variances $(\mathrm{var}(x_1), \mathrm{var}(x_2)) = (\sigma_1^2, \sigma_2^2)$, with $\sigma_1^2 > \sigma_2^2$. Set $K = 2$, assume N is large, and investigate the fixed points of the algorithm as β is varied. [Hint: assume that $m^{(1)} = (m, 0)$ and $m^{(2)} = (-m, 0)$.]

Solution:

Exercise 20.4

Consider the soft K-means algorithm applied to a large data set in which the points can be modelled as a mixture of narrow and broad clusters. What behaviour does the algorithm show as β is increased and why?

Solution:

Exercise 20.5

Why does the pitchfork in figure 20.11 tend to the value $m = 0$ when the stiffness parameter is small? Explain the mechanism qualitatively.

Solution: